

---

# Lab 1: Transfer Data to S3 Using Amazon Kinesis Data Firehose

## Objective:

Ingest real-time streaming data and deliver it to an S3 bucket using **Amazon Kinesis Data Firehose**.

---

## Prerequisites:

- AWS Account with IAM access
  - IAM Role for Firehose with access to S3 and (optionally) Lambda
  - S3 bucket to store the data
  - (Optional) AWS CLI or SDK for testing
- 

## Step 1: Create an S3 Bucket

```
aws s3 mb s3://my-firehose-demo-bucket
```

Or via AWS Console:

- Go to S3 → Create Bucket → Enable versioning (optional) → Create
- 

## Step 2: Create IAM Role for Firehose

Attach the following policy to allow access:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetBucketLocation"
      ],
      "Resource": [
        "arn:aws:s3:::my-firehose-demo-bucket",
        "arn:aws:s3:::my-firehose-demo-bucket/*"
      ]
    }
  ]
}
```

Create IAM Role with:

- **Service:** firehose.amazonaws.com
- Attach this policy



## Step 3: Create a Firehose Delivery Stream

- Go to Amazon Kinesis → Firehose → Create delivery stream
  - Source: **Direct PUT**
  - Destination: **Amazon S3**
  - S3 bucket: **my-firehose-demo-bucket**
  - Buffer conditions (e.g., 1MB or 60 seconds)
  - Choose or create IAM Role
-

## Step 4: Send Data to Firehose Stream

### Using AWS CLI:

```
aws firehose put-record \  
  --delivery-stream-name my-firehose-stream \  
  --record='{ "Data": "{ \"user_id\": 123, \"action\": \"click\", \"timestamp\":  
  \"2025-07-29T10:00:00Z\"}" }
```

### Sample Output:

```
{  
  "RecordId": "Dzr1oEXAMPLER/0A6k1pQ=="  
}
```

---

## Step 5: Verify Output in S3

Go to S3 bucket:

- Check for files in the prefix (time-based)
  - Download and open the delivered file
- 

## Done!

You've now successfully used Kinesis Firehose to ingest data and deliver it to S3.

---

## Lab 2: Amazon Kinesis Data Streams (with Lambda Consumer)

---

### Objective:

Stream real-time JSON data into Kinesis Data Streams and process it using **AWS Lambda**, storing results in S3.

---

## ✓ Step 1: Create a Kinesis Data Stream

```
aws kinesis create-stream --stream-name my-kinesis-stream --shard-count 1
```

Or Console:

- Go to Amazon Kinesis → Data Streams → Create Stream
  - Name: `my-kinesis-stream`
  - Shards: 1
- 

## ✓ Step 2: Create a Lambda Function

- Go to Lambda → Create Function
- Runtime: Python or Node.js
- Permissions: Add policy for S3 PutObject access
- Sample handler:

```
import json
import boto3
import uuid
```

```
s3 = boto3.client('s3')
```

```
def lambda_handler(event, context):
    for record in event['Records']:
        payload = json.loads(record['kinesis']['data'])
        key = f"record-{uuid.uuid4()}.json"
        s3.put_object(Bucket="my-kinesis-lambda-bucket", Key=key, Body=json.dumps(payload))
    return {'status': 'done'}
```

---

### ✓ Step 3: Create S3 Bucket for Output

```
aws s3 mb s3://my-kinesis-lambda-bucket
```

---

### ✓ Step 4: Configure Lambda Trigger

- Go to Lambda → Add Trigger
  - Choose Kinesis
  - Select Stream: `my-kinesis-stream`
  - Batch size: 100
  - Enable Trigger
- 

### ✓ Step 5: Put Data into the Stream

```
aws kinesis put-record \  
  --stream-name my-kinesis-stream \  
  --partition-key 1 \  
  --data "$(echo -n '{"sensor_id": 1001, "value": 45}' | base64)"
```

---

### ✓ Step 6: Check Data in S3

- Go to S3 → `my-kinesis-lambda-bucket`
  - Download one of the JSON records to verify
-

## Summary

| Lab                     | Description                       | Output                           |
|-------------------------|-----------------------------------|----------------------------------|
| <b>Firehose to S3</b>   | Real-time delivery with no coding | S3 file delivery                 |
| <b>Streams + Lambda</b> | Custom logic with transformation  | S3 structured data, alerts, etc. |

---